

HONG KONG BAPTIST UNIVERSITY

COURSE OUTLINE

1. COURSE TITLE

Financial Technology

2. COURSE CODE

COMP7300

3. NO. OF UNITS

3 Units

4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

5. PREREQUISITES

Postgraduate Student Standing

6. MEDIUM OF INSTRUCTION

E

7. AIMS & OBJECTIVES

To provide students with an in-depth understanding of IT applications and latest technological developments in the financial industry. Students will gain an understanding of the operations in the financial sector, as well as the latest technologies adopted in this field. Practical elements will also be included to allow students to experience the use and development of technologies to support the operations and decision making of financial processes.

8. COURSE CONTENT

I. Introduction to Financial Industry and operations

A. Financial concepts

- Governance and regulatory frameworks
- Global FinTech trends
- Compliance and risk management for FinTech
- Traditional financial services, such as deposit taking and lending, cheque processing, remittance, international trade, credit card processing, trading and market surveillance
- Emerging FinTech applications, such as digital payments systems, digital currency,

robo-advisor, cybersecurity

II. Technologies Adopted in Financial Industry

A. Traditional financial applications

- Financial electronic communication networks
- Payment processing systems
- Clearing and settlement systems
- Financial decision support systems

B. Disruptive technologies

- Artificial intelligence, machine learning and deep learning
- Blockchain technologies
- Cybersecurity technologies
- Cloud Computing
- Financial Data Engineering
- Other emerging technologies used in the industry

III. Development of Financial Applications

A. Database design

B. Prototyping

C. Agile development methodologies

9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

CILO	By the end of the course, students should be able to:
CILO 1	Explain the basic concepts and operations in the financial industry
CILO 2	Justify the latest technologies used in the financial sector
CILO 3	Demonstrate the skills to develop financial applications
CILO 4	Evaluate the functions of selected financial applications
CILO 5	Implement an application to support financial operations

10. TEACHING & LEARNING ACTIVITIES (TLAs)

CILO alignment	Type of TLA
1-3	Students will attend lectures for the financial operations. They will also gain knowledge of the up-to-date technologies used in the financial industry, and development skills to implement financial applications.
4	Students will practice selected financial applications via laboratory session.
5	Students will work on a project to apply the development skills for financial application implementation.

11. ASSESSMENT METHODS (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Continuous assessment	50 %	1-5	Continuous assessments measure students' understanding of technology applications in the financial industry and the ability to manipulate the functions of current financial applications. A project will evaluate students' ability to develop a financial application.
Examination	50 %	1-3,5	Final examination questions are designed to evaluate how far students have achieved their intended learning outcomes.

Last Update: 2025-10-27
Published Date: 2025-11-05

***** END *****